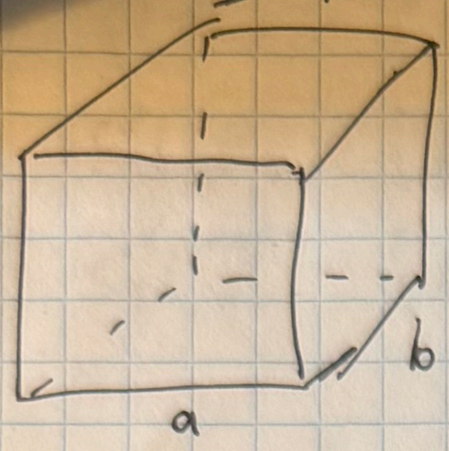
Solid State Physics HW1

Exercise 1: Free Electrons in a Box



$(a + b + c)$

$a, b, c \to \infty$

$T = 0$

a) Single particle, empty cube $\Rightarrow V(x) = 0$, $T = 0$, TISE:

$$\Rightarrow -\frac{\hbar}{2m} \nabla^2 \psi(\underline{r}) = E \psi(\underline{r})$$

Boundary conditions:

$$\psi(x+a, y, z) = \psi(x, y, z)$$
$$\psi(x, y+b, z) = \psi(x, y, z)$$
$$\psi(x, y, z+c) = \psi(x, y, z)$$

Normalized plane waves: $\psi_{\underline{k}}(\underline{r}) = e^{i \underline{k} \underline{r}} / \sqrt{V}$

$$-\frac{\hbar}{2m} \nabla^2 \psi_{\underline{k}}(\underline{r}) = -\frac{\hbar}{2m} \frac{1}{\sqrt{V}} \nabla^2 e^{i(k_x x + k_y y + k_z z)} = -\frac{\hbar}{2m\sqrt{V}} \left((ik_x)^2 + (ik_y)^2 + (ik_z)^2 \right) e^{i \underline{k} \underline{r}}$$

$$= -\frac{\hbar}{2m\sqrt{V}} \left(-(k_x^2 + k_y^2 + k_z^2) \right) e^{i \underline{k} \underline{r}} = -\frac{\hbar}{2m\sqrt{V}} (-k^2) e^{i \underline{k} \underline{r}} = \underbrace{\frac{\hbar k^2}{2m}}_{\hookrightarrow \text{constant}} \psi_{\underline{k}}(\underline{r})$$

$\Rightarrow \psi_{\underline{k}}(\underline{r})$ are eigenstates

Allowed wavevectors:

boundary on x-axis: $\psi_{\underline{k}}(x+a, y, z) = \psi_{\underline{k}}(x, y, z)$

$$\Rightarrow e^{i(k_x(x+a) + k_y y + k_z z)} = e^{i(k_x x + k_y y + k_z z)} \quad \Rightarrow e^{i k_x a} = 1$$

$$\Rightarrow i k_x a = i 2\pi n_x \quad (n_x \in \mathbb{Z}) \quad \Rightarrow k_x = \frac{2\pi n_x}{a}$$

similarly on y, z: $k_y = \frac{2\pi n_y}{b}$ $\quad k_z = \frac{2\pi n_z}{c}$ $\quad (n_x, n_y, n_z \in \mathbb{Z})$

$$\Rightarrow \underline{k} = \left(\frac{2\pi n_x}{a}, \frac{2\pi n_y}{b}, \frac{2\pi n_z}{c} \right)$$

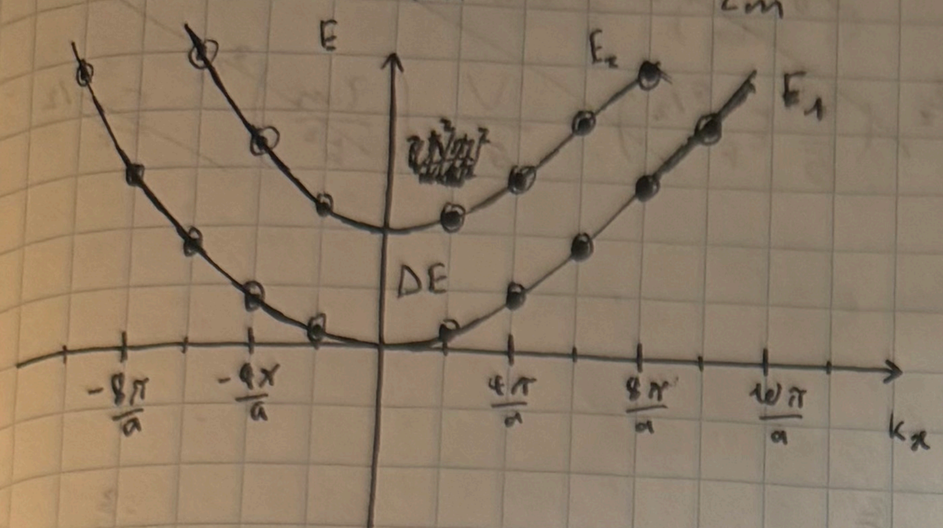
Volume of k-space by single allowed $\underline{k}$-point:

$$\Delta V_k = \Delta k_x \Delta k_y \Delta k_z = \frac{2\pi}{a} \cdot \frac{2\pi}{b} \cdot \frac{2\pi}{c} = \frac{8\pi^3}{abc} = \frac{8\pi^3}{V}$$

Electron states (w/spin) in volume Ω_k k-space:

$$\text{\# Electron states} = 2 \cdot \text{\# k-points} = 2 \frac{\Omega_k}{\Delta V_k} = \frac{\Omega_k V}{4\pi^3}$$

b) $E(k_x) = \frac{\hbar^2 k_x^2}{2m} + \frac{\hbar^2(k_y^2 + k_z^2)}{2m}$



$(k_y, k_z) = (0,0)$: $E_1(k_x) = \frac{\hbar^2 k_x^2}{2m}$ +

$(k_y, k_z) = (\frac{2\pi}{b}, 0)$: $E_2(k_x) = \frac{\hbar^2 k_x^2}{2m} + \frac{2\hbar^2\pi^2}{mb^2}$

$\Delta E = \frac{2\hbar^2\pi^2}{mb^2}$

states with same degeneracy: $\{k_x, -k_x\}$
(and spin degen. for each state, 2 from $|\uparrow\rangle, |\downarrow\rangle$)

a increase:
⇒ $\Delta k_x = \frac{2\pi}{a}$ decrease
⇒ more possible energy state per energy interval
⇒ density of states increase

c) Number of states $\mathcal{N}(E)$ below E:

$E = \frac{\hbar k^2}{2m}$ (from a)) ⇒ $k = \sqrt{\frac{2mE}{\hbar}} = k_{max}$

Volume of sphere in k-space (Fermi sphere): $\Omega_k = \frac{4}{3}\pi k^3$

⇒ $\mathcal{N}(E)$ = # electron states = $\frac{\Omega_k V}{4\pi^3}$ (from a)

$= \frac{V}{3\pi^2} k^3 = \frac{V}{3\pi^2}\left(\sqrt{\frac{2mE}{\hbar}}\right)^3 = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} E^{\frac{3}{2}}$

↑ $k = k_{max}$

Density of states: $g(E)$

$g(E) = \frac{d\mathcal{N}(E)}{dE} = \frac{d}{dE}\left(\frac{V}{3\pi^2}\left(\sqrt{\frac{2m}{\hbar}}\right)^3 E^{\frac{3}{2}}\right) dE = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \frac{3}{2}\sqrt{E}$

⇒ $g(E) \sim V = a \cdot b \cdot c$ = shape (a, b, c) of sample

d) N electrons, $T = 0$:

$\mathcal{N}(E_F) = \frac{V}{3\pi^2} k_F^3$ (part c) $= N$ (N electrons fill lowest states → k_F energy)

⇒ $k_F = (3\pi^2 \frac{N}{V})^{\frac{1}{3}} = (3\pi^2 n)^{\frac{1}{3}}$

⇒ $E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{\frac{2}{3}}$ $(E = \frac{\hbar^2 k^2}{2m})$

$g(E_F) = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \frac{3}{2}\sqrt{E_F}$ $= \frac{\mathcal{N}(E_F)}{E_F^{\frac{3}{2}}} \frac{3}{2}\sqrt{E_F}$ (part c)

$= N \frac{3}{2\sqrt{E_F}} = \frac{3N}{2E_F}$ (pow ⇒ shown

$E_0 = \int_0^{E_F} E\, g(E)\, dE = \int_0^{E_F} E\left(\frac{3N}{2} E_F^{-3/2} E^{1/2}\right) dE$

$= \frac{3N}{2} E_F^{-3/2} \int_0^{E_F} E^{3/2} dE = \frac{3}{5} N E_F$

$N = 2.5 \times 10^{21}$ electrons, $V = 2\,mm \times 3\,mm \times 5\,mm = 30 \times 10^{-9}\,m = 30 \times 10^{21}\,Å^3$

⇒ $n = \frac{N}{V} = 1/12\ Å^{-3}$

⇒ $k_F = (3\pi^2 n)^{1/3} \approx 1.351\ Å^{-1}$

⇒ $E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3} \approx 6.956\ eV$

⇒ $T_F = E_F / k_B \approx 80724\,K$

e) i) $E_0 = \frac{3}{5} N E_F$ (part d) $= \frac{3}{5} n V E_F$

$E_0 = \int_0^{E_F} E\, g(E)\, dE = \int_0^{E_F} E \left(\frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{E}\right) dE$

$= \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \left(\frac{2}{5} E_F^{5/2}\right) = \frac{V}{5\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{5/2}$

$P = -\frac{\partial E_0}{\partial V} = \frac{1}{5\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{5/2} \frac{\partial}{\partial V}(V)$

e) i) $E_0 = \frac{3}{5} N E_F$ (part d)

$= \frac{3}{5} N \frac{\hbar^2}{2m} (3\pi^2 \frac{N}{V})^{2/3}$ $\propto V^{-2/3}$ $\propto V^{-\frac{2}{3}}$

$P = -\frac{\partial E_0}{\partial V} = -\left(-\frac{2}{3} \frac{E_0}{V^{+2/3}} V^{-\frac{5}{3}}\right) = \frac{2}{3}\frac{E_0}{V} = \frac{2}{3}\frac{3}{5} n E_F = \frac{2}{5} n E_F$ $\propto \frac{V^{-\frac{3}{3}}}{V} = V^{-\frac{5}{3}}$

$B = -V \frac{\partial P}{\partial V} = -V\left(-\frac{5}{3}\frac{P}{V}\right) = \frac{2}{3} n E_F$

ii) Compress x: $a \to \lambda a$ $(\lambda < 1)$; b, c unchanged ; (n_x, n_y, n_z) unchanged

$\langle k_x^2 \rangle = \frac{k_F^2}{5} = \langle k_y^2 \rangle = \langle k_z^2 \rangle$ (sphere)

To prove: $E_{ad}(\lambda) = \frac{1}{3} E_0 (\lambda^{-2} + 2)$

$a \to \lambda a \Rightarrow k_x \to 2\pi n / \lambda a = k_x / \lambda$, k_y, k_z unchanged

$E = \frac{\hbar^2}{2m} \sum \underline{k}^2 = \frac{\hbar^2}{2m} N \left(\langle k_x^2 \rangle + \langle k_y^2 \rangle + \langle k_z^2 \rangle\right)$

$\Rightarrow E_{ad} = \frac{\hbar^2}{2m} N \left(\frac{\langle k_x^2 \rangle}{\lambda^2} + \langle k_y^2 \rangle + \langle k_z^2 \rangle\right)$

$= \frac{\hbar^2}{2m} N \left(\frac{1}{\lambda^2}\frac{k_F^2}{5} + \frac{k_F^2}{5} + \frac{k_F^2}{5}\right) = \frac{\hbar^2}{2m} N \frac{k_F^2}{5} (\lambda^{-2} + 2)$

Before compression $(\lambda = 1)$: $E_0 = \frac{3}{5} N E_F = \frac{3}{5} N \frac{\hbar^2 k_F^2}{2m}$

$\Rightarrow E_{ad} = \frac{1}{3} E_0 (\lambda^{-2} + 2)$ $\Rightarrow$ proven

$P_x = \frac{F}{A} = \frac{F_x}{bc} = \frac{1}{bc}\frac{\partial E}{\partial a}$ $\lambda = \frac{a}{a_0}$

$P_x = -\frac{\partial E}{\partial V} = -\frac{\partial E_{ad}}{bc\, \partial a} = -\frac{1}{bc}\frac{\partial}{\partial a}\left(\frac{1}{3} E_0 \left(\frac{a_0^2}{a^2} + 2\right)\right) = -\frac{1}{bc}\left(-\frac{2E_0}{3}\frac{a_0^2}{a^3}\right)$

at $\lambda = 1 \Rightarrow a = a_0$:

$\Rightarrow P_x = -\frac{1}{bc}\left(-\frac{2E_0}{3a_0}\right) = \frac{2}{3}\frac{E_0}{V} = \frac{2}{3}\frac{3}{5}\frac{N}{V} E_F = \frac{2}{5} n E_F$ = i) same

iii) E_{ad} compare to E_{rel}

$E_0 \propto V^{-\frac{2}{3}}$ (part i) $\Rightarrow E_{rel} = E_0 \left(\frac{V'}{V}\right)^{-2/3} = E_0 \lambda^{-2/3}$

$\lambda = 1 - \varepsilon$, $\varepsilon \ll 1$: $(1-\varepsilon)^{-p} = 1 + p\varepsilon + \frac{p(p+1)}{2}\varepsilon^2 + \dots$

$\Rightarrow E_{ad} = \frac{1}{3} E_0 (\lambda^{-2} + 2) = \frac{1}{3} E_0 [3 + 2\varepsilon + 3\varepsilon^2] = E_0 \left(1 + \frac{2}{3}\varepsilon + \varepsilon^2\right)$

$\Rightarrow E_{rel} = E_0 \left(1 + \frac{2}{3}\varepsilon + \frac{5}{9}\varepsilon^2\right)$

$E_{ad} - E_{rel} = E_0 \left(\varepsilon^2 - \frac{5}{9}\varepsilon^2\right) = \frac{4}{9} E_0 \varepsilon^2$ (part c)

$n = \frac{1}{12}\ \text{Å}^{-3}$ (part d), $\varepsilon^2 = 10^{-3}$, $E_0 = \frac{3}{5} N E_F$, $E_F = 6.956\ \text{eV}$

$\Rightarrow \frac{E_{ad} - E_{rel}}{N} = \frac{4}{9} \cdot \frac{3}{5} \frac{N}{N} E_F \varepsilon^2 \approx 1.9 \times 10^{-6}\ \text{eV}$

Shape of occupied region:

$k_x^2 + k_y^2 + k_z^2 \leq k_F^2 \xrightarrow{\text{compress}} \lambda^2 k_x'^2 + k_y'^2 + k_z'^2 \leq k_F^2$

$\Rightarrow \frac{\lambda^2}{k_F^2} k_x'^2 + k_y'^2 + k_z'^2 \leq 1$ $\Rightarrow$ ellipsoid